

Trainings ▾

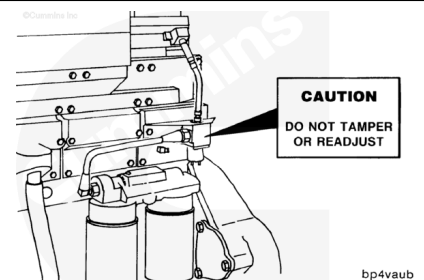
General Information

Some engine models are equipped with step timing control (STC). Step timing control allows the engine to operate in advanced injection timing immediately after start-up and light duty engine load conditions, and to return to normal timing during medium and high engine load conditions.

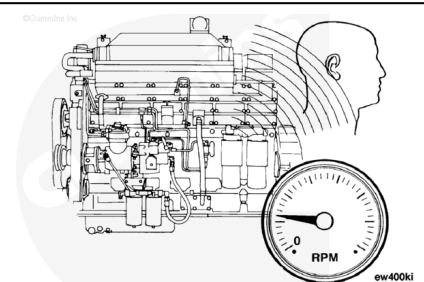
Benefits include:

- Improved cold weather idling characteristics
- Reduced cold weather white smoke
- Improved light load fuel economy.

Do **not** attempt to bypass or otherwise tamper with the STC oil control valve or plumbing. This will result in the loss of both fuel economy and engine durability. Correct valve operation is necessary to maintain acceptable cylinder pressures and temperatures, and to yield optimal fuel economy during high-load operation. Correct operation is also necessary to control white smoke at idle.



When operating in the advanced mode, a light ticking noise can be noted at the overhead. This sound is normal, and is caused by the actuation of the STC hydraulic tappets during each injection cycle.



For optimal white smoke control on STC-equipped engines, do not increase engine speed above idle during engine start-up until sufficient oil pressure reaches the STC tappets and shifts all injectors into the advanced timing mode.

